Design of Contract Management System Based on J2EE Architecture

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***Abstract*—With the rapid development of information technology, E-Business system has gradually become an effective means for enterprises to raise their management efficiency, save office resources, and achieve reasonable resource allocation. The traditional contract management mechanism for most enterprises is completed by manual processing, thus, the official business efficiency is very low and the waste of resource is serious. Therefore, this paper presents Design of Project Contract Management System Based on J2EE Architecture to solve the existing contract management problems. The system design follows the principles of advanced, economical, reliable, high efficient, open, easy to operate and easy to maintenance. According to the characteristics of projects, the system design is based on B/S structure, J2EE platform, MVC model design scheme. Based on the UML system analysis methods, we carry out a detail analysis on the contract management's relative business and practical functions that need to be realized. On the basis of the feasibility study and requirement analysis, this paper discusses the system function design and the database design in detail. By the way, we carry out the system functions by using the J2EE integration architecture based on WebWork + Spring + Hibernate. The contract management system provides a scientific and highly efficient contract management platform for the majority of enterprises.**

Keywords-contract management; design; J2EE; MVC; Webwork+ Spring+Hibernate

I. INTRODUCTION

Contract management is an indispensable part in the enterprise's business. Raising contract management's efficiency has positive significance for the enterprises economic activities and economic benefits made. Contract management running in good condition has benefit to the enterprise internal resources optimization allocation.

The contract management system is based on information technology, it is an omnidirectional systematic contract management system that provides decision, plan, control and business performance evaluation for the enterprise based on all kinds of data the enterprises provide. A perfect contract management system is an important sign of corporates' healthy operation. A perfect contract management system, however, requires many resources, how to simplify the management of the contract without losing its integrity and science is a headache problem for many business.

This paper mainly discusses the Design of Project Contract Management System Based on J2EE Architecture. This contract management system is an application system that provides service for the projects based on the database programming. This paper has elaborated the contract management system design idea and the system function architecture.

II. SYSTEM ARCHITECTURE

A. B/S

B/S architecture (Browser/Server) is a network model with the growing up of Web, and the Web Browser is one of the most important applications of client [1].

After such cross-platform language Java appeared , it's more convenient, faster and more efficient for the B/S structure to manage software.

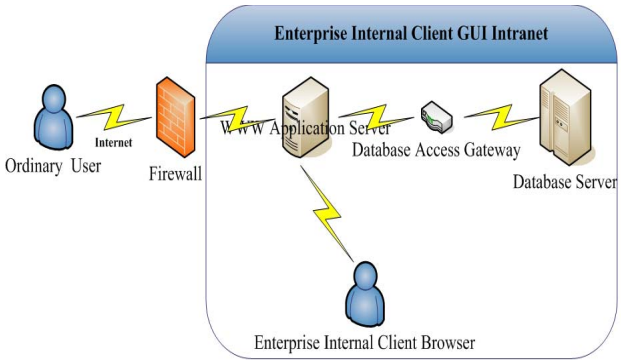

Figure 1. B/S structure schematic diagram

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B. J2EE and MVC

J2EE is a de facto industry standard that uses Java technology to develop enterprise application [2]. The purpose of introducing J2EE is to overcome the malady of traditional Client/Server Mode, to cater to the trend of the Browser / Server structure, and to provide a platform independent, portable, multi-user, security and based on standard enterprise-class platform for using Java technology to develop server side applications, which simplifies the development, management and deployment of enterprise applications.

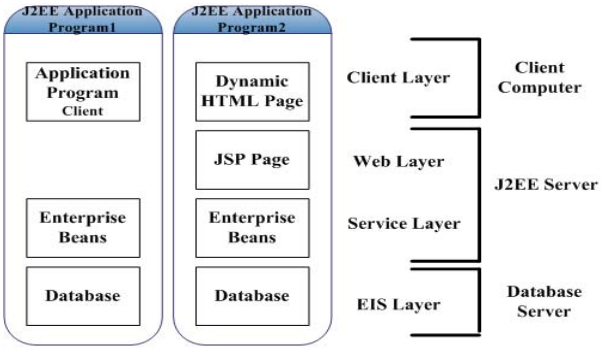

Figure 2. J2EE four layers structure

MVC, M refers to model, V is view, and C is controller. The purpose of using the MVC pattern is to realize code's separation of Model and View, which makes the same program can be used in different forms of expression. The purpose of existence of controller is ensure model and view synchronization, once model change, view should update [4].

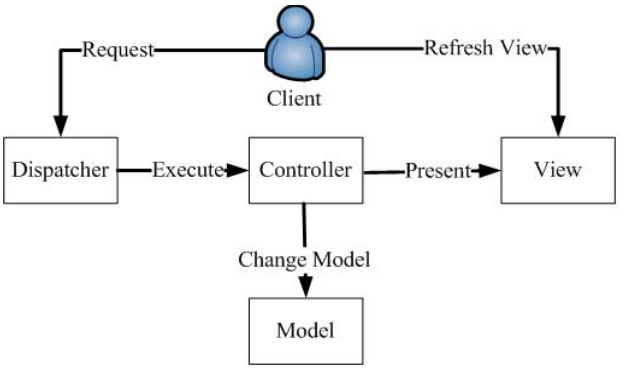

Figure 3. The Principle of MVC Pattern

C. Webwork+Spring+Hibernate

The WebWork+Spring+Hibernate basic structure based on J2EE can be divided into three parts: that is, presentation layer, business logic layer, data persistence layer. Among them: the presentation layer also called Web layer, J2EE Web component can be made up by the components such as the JSP pages; Business layer also called EJB layer, to perform as to meet a specific business need in the field of business logic code from running in the EJB business layer; Data persistence layer is the layer of database data management, mainly responsible for storaging business data persistently.

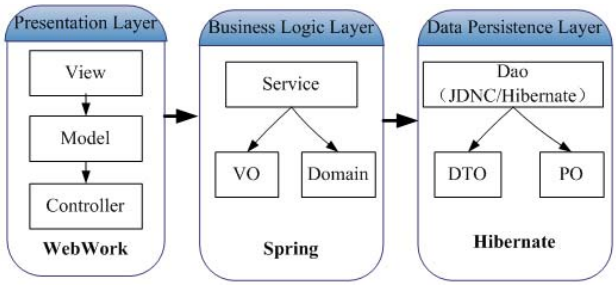

Figure 4. System call three layer structure model diagram

During the development process, the three kinds of framework technology have the steady development background and extensive documentation; The integrated framework makes each layer affect each other in a way of loosely coupling so that we can ignore the underlying technical details. All the three frameworks integrated, can make different frame in minimum limitations, and make full use of the advantages of each frame to achieve good effect.

III. CONTRACT MANAGEMENT SYSTEM REQUIREMENT ANALYSIS

A. System Requirement Analysis

1) User profile

Contract management system described in this paper is mainly to be used to contract information management, the auxiliary function have project information management, company information management, contract type information management, etc. The user of the system is the enterprises' office personnel.

2) System target

- Strengthen the scientific of contract information management, and improve the processing efficiency of contract information.
- Centralized data management, enhance data sharing.
- Reduce using difficulty, to provide a more convenient operating mode: (1) Pages are concise and elegant and the operating processes are simple, tedious process to achieve the easy steps to facilitate the no experience user to operate the information management system; (2) Have statistical and query functions, users can not only see all kinds of information in the system but also query the contract information, project information, company information in various combinations.
- Clear responsibilities and authority, regulate the scope of management.
- Provide function expansion interface.

3) Contract management system organization structure

Based on the requirement analysis of projects, the contract system in general is divided into five major modules, it is respectively: contract information management, project information management, company information management, contract type management, system management. Overall function structure shown as follows:

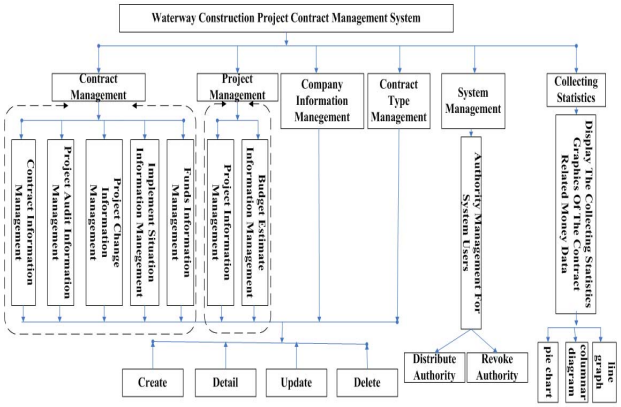

Figure 5. Overall function structure

4) Function requirement analysis

According to the requirements of users, the contract system is to validate and manage contracts information, contract management module is located in the central position of the entire contract management system, but contract information processing will also rely on the information provided by project information management, company information management, contract type management, etc. At the same time, the contract information management has five sub processings: engineering audit information management, project change management, implementation information management, funds information management; And project information management including sub processing: budget management.

B. System Business Process Analysis

1) Use case diagram

Use case diagram describes the functional unit the system provides. In the system analysis stage, we can use the use case diagram to describe the system business process and system logic model.

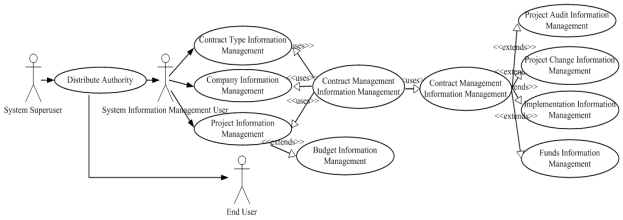

Figure 6. Overall user case diagram

2) Swimlane diagram

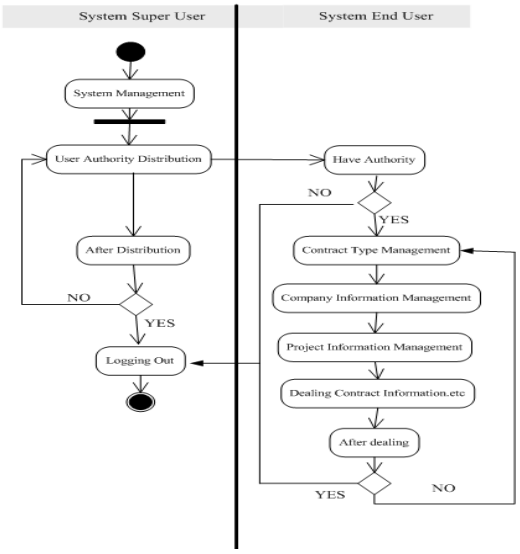

Figure 7. Contract system swimlanes diagram

Swimlane diagram organizes the activities in the model by responsibilities [3].

IV. CONTRACT MANAGEMENT SYSTEM DESIGN

A. System Design Principal

To construct a successful system, should not only need to consider how to accomplish system functions in an effective and efficient way, but also make the system extensible, in a certain period should adapt to the development and function expansion of the requirements, and at the same time to provide the end users with convenient and practical information service.

B. System Function Design

The contract system class diagram is shown below.

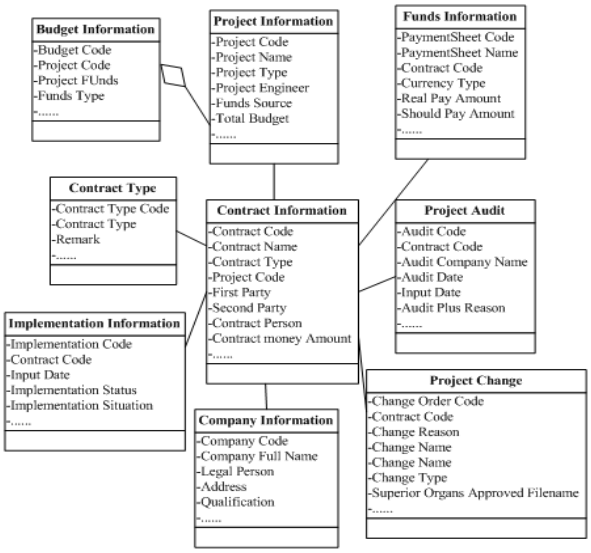

Figure 8. Contract system function structure class diagram

C. Database and Data Dictionary Design

1) Database Design Principle

On the basic of previous analysis, the data the system users need to use is basically understood by us, then it should carry on a detailed analysis of the data to establish the data model.

During the database design, should follow the following several principles: (1)The general principles; (2) The consistency principle; (3)The security principle [5].

2) Database Design

Subject database design primarily for contract management business design. The other system control, configuration table are omitted, the following is the subject of the contract management system database.

TABLE I. SUBJECT DATEBASE

Subject database sign	***Subject database name***	***Database table sign***	***Database table name***
contract_user	Contract System Database	t_change_order	Project Change Table
		t_contractinfo	Contract Table
		t_cost_type	Budget Table
		t_implementation	Implementation Table
		t_organizationinfo	Company Table
		t_project_audit	Project Audit Table
		t_projectinfo	Project Information Table
		t_single_payment	Funds Table
		t_type	Contract Type Table
		t_access	Access Control Table
contract_basic	Basic Database	t_user	User Information Table

3) Data Dictionary Design

Take contract information table (T_CONTRACTINF) as an example. Part of the field data dictionary design as follows:

TABLE II. CONTRACT INFORMATION TABLE

Name	***Code***	***Date Type***	***Primary***	***Foreign Key***
ID	ID	INTEGER	True	False
contract name	HTMC	VARCHAR（100）	False	False
……	……	……	……	……
remaining currency amount	SYJE	FLOAT（10）	False	False
tentative total currency amount	ZDJEZJ	FLOAT（10）	False	False

V. CONCLUSION

With the rapid development of network technology, more and more application system based on Web coming into system developer's vision, through the completion of the contract management system, the contract management must be more systematic and more standardized, it can improve the efficiency of management and reduce the cost of contract management, this study is the focus on here.

ACKNOWLEDGMENT

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REFERENCES

[1] Deng XinXin, Qiu ZhongPan and Yang Xiaofang. Research and design of network behavior management system based on B/S Architecture[J], The 6th International Conference on Computer Science & Education (ICCSE 2011). August 3-5, 2011.

[2] http://wenku.baidu.com/view/470b1181d4d8d15abe234e0e.html

[3] Guo Gang, Zhou Feng and Sun Gengxin. UML and Rational Rose 2003 software engineering unified modeling principle and practice tutorial[M].Beijing: Electronic Industry Press. April, 2007

[4] David Lowe,Brian Henderson-Sellers,Alice Gu.Web extensions to UML:Using the MVC Triad.Conceptual Modeling-ER 2002[M],2002:74-93.

[5] Chen Jia. Information system development method course (The Second edition)[M]. Beijing: Tsinghua Publishing House, 2005.